

Roll No.....

MID SEMESTER EXAMINATION MARCH - 2025

Name of the Program: B.Tech. (ME-RA)

Semester: **VIII**

Name of the Course: Automatic Control

Course Code: **TRA 801**

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define the following terms used in control systems: i) Plant, ii) Manipulated variables, iii) Controlled variables, iv) Feedback control	
	OR	
(b)	What do you understand by the open-loop and closed-loop control systems? Discuss with their respective advantages.	CO1
Q2	(10 marks)	
(a)	What do you understand by the Poles and Zeros of a system? Discuss about the stability of a system based on the Poles mapping on s-plane.	CO2
	OR	
(b)	Plot and write the expressions for the following signals: step, ramp, sinusoidal, and impulse.	CO2
Q3	(10 marks)	
(a)	Find the solution $x(t)$ of the following differential equation: $\ddot{x} + 2\dot{x} + 5x = 3$; where $x(0) = 0$ and $\dot{x}(0) = 0$	CO2
	OR	
(b)	Find the inverse Laplace transform of the following function: $F(s) = \frac{s + 4}{(s + 2)(s + 3)}$	CO2
Q4	(10 marks)	
(a)	What do you understand by the Routh's stability criterion? Write the different steps of the Routh's stability criterion.	CO3
	OR	
(b)	Comment about the stability of a system using Routh's stability criterion, whose characteristic equation is given by the following polynomial: $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$	CO3
Q5	(10 marks)	
(a)	Derive the expression for the unit-step response of a first-order system.	CO3
	OR	
(b)	What do you understand by the Transient response, Steady-state response, and Steady-state error?	

